

# Question 1: Hospital Patient Segmentation

## 1.1 Dataset & Given Parameters

| Patient ID | Age | Monthly Medical Expense (₹) | Health Risk Score |
|------------|-----|-----------------------------|-------------------|
| P1         | 28  | 12,000                      | 65                |
| P2         | 55  | 25,000                      | 40                |
| P3         | 42  | 20,000                      | 75                |
| P4         | 30  | 11,000                      | 70                |
| P5         | 60  | 28,000                      | 35                |

| Number of Clusters (K) | Sum of Squared Errors (SSE) | $\Delta$ SSE (Reduction from K-1) | Marginal % Improvement |
|------------------------|-----------------------------|-----------------------------------|------------------------|
| K = 1                  | 600                         | -                                 | -                      |
| K = 2                  | 350                         | 250                               | 41.67%                 |
| K = 3                  | 220                         | 130                               | 37.14%                 |
| K = 4                  | 200                         | 20                                | 9.09%                  |
| K = 5                  | 190                         | 10                                | 5.00%                  |

## 1.2 Interpretation of the Elbow Point & Optimal Number of Clusters

**Elbow Method Principle:** The Elbow Method evaluates the Sum of Squared Errors (SSE) as a function of the number of clusters (K) to identify where the rate of decrease sharply diminishes (the 'elbow'). SSE represents the total intra-cluster variance, measuring how tightly grouped points are around their respective cluster centroids.

- **Transition from K=1 to K=2:** SSE drops substantially from 600 to 350 ( $\Delta$  = 250, 41.7% reduction).
- **Transition from K=2 to K=3:** SSE drops significantly from 350 to 220 ( $\Delta$  = 130, 37.1% reduction).

- **Transition from K=3 to K=4:** SSE drops only marginally from 220 to 200 ( $\Delta = 20$ , 9.1% reduction).
- **Transition from K=4 to K=5:** SSE further plateaus to 190 ( $\Delta = 10$ , 5.0% reduction).

**Conclusion on Optimal K:** The inflection point ('elbow') clearly occurs at  $K = 3$ . Moving beyond  $K = 3$  yields diminishing marginal returns in error reduction while increasing model complexity and risking overfitting. Hence,  $K = 3$  is mathematically and operationally the optimal number of clusters.

1.3 Patient Segmentation Profiles & Healthcare Decision-Making

Based on  $K = 3$ , the patient dataset naturally segments into three distinct clinical and economic profiles:

- **Cluster 1 — Young, Low Expense, Moderate/High Risk (P1, P4):** Average age ~29, monthly expenses ~₹11,500, risk scores 65–70. These individuals are early-stage or preventive-care candidates.  
Decision-Making: Implement proactive health wellness apps, lifestyle counseling, routine screening, and affordable preventive insurance plans to avoid disease escalation.
- **Cluster 2 — Older, High Expense, Moderate Risk (P2, P5):** Average age ~57.5, high monthly expenses ~₹26,500, risk scores 35–40. These patients likely suffer from managed chronic conditions (e.g., hypertension, arthritis) requiring continuous medication. Decision-Making: Optimize chronic disease management, pharmacy bulk subsidies, home delivery of maintenance medications, and specialized geriatric care coordination.
- **Cluster 3 — Middle-Aged, High Risk, Moderate-to-High Expense (P3):** Age 42, expenses ₹20,000, risk score 75 (highest risk). This patient represents an acute/critical vulnerability. Decision-Making: Immediate clinical intervention, personalized specialist care, dedicated nursing support, and intensive monitoring to mitigate immediate catastrophic medical emergencies.

Question 2: E-Commerce Order Analysis

2.1 Dataset & Order Characteristics

| Order ID | Product Category | Order Value (₹) |
|----------|------------------|-----------------|
| O1       | Electronics      | 55,000          |
| O2       | Clothing         | 25,000          |
| O3       | Electronics      | 57,000          |
| O4       | Furniture        | 30,000          |
| O5       | Clothing         | 27,000          |

## 2.2 Hierarchical Clustering & Dendrogram Interpretation

**Hierarchical Merging Process:** Agglomerative Hierarchical Clustering builds a bottom-up tree (dendrogram) by iteratively merging the most similar orders based on both categorical compatibility and numerical proximity (Order Value in ₹):

- **Step 1 (First Merge):** Orders O2 (Clothing, ₹25,000) and O5 (Clothing, ₹27,000) are merged first at the lowest vertical distance ( $\Delta\text{Value} = ₹2,000$ , identical category).
- **Step 2 (Second Merge):** Orders O1 (Electronics, ₹55,000) and O3 (Electronics, ₹57,000) are merged next at a small distance ( $\Delta\text{Value} = ₹2,000$ , identical category).
- **Step 3 (Third Merge / Sub-Branch):** Order O4 (Furniture, ₹30,000) is close in order value to the {O2, O5} group (distance  $\sim ₹4,000$  from mean), forming a broader mid-value tier or remaining an independent category branch.
- **Step 4 (Final Merge):** The high-value cluster {O1, O3} merges with the lower/mid-value cluster {O2, O4, O5} at a very high vertical threshold (distance  $> ₹25,000$ ).

**Optimal Number of Clusters:** By cutting the dendrogram horizontally across the longest vertical branches (where distance jump is maximum between ₹30,000 and ₹55,000), the optimal number of clusters is identified as either  $K = 2$  (Macro-segmentation by Value Tier) or  $K = 3$  (Domain-specific Segmentation by Category & Value).

## 2.3 Significance in Customer Purchasing Behavior

- **Cluster A — High-Ticket Tech Consumers {O1, O3}:** Customers purchasing high-value Electronics (₹55k–₹57k). Behavior: High spending capacity, deliberate planned purchases, sensitivity to warranties, tech specifications, and flexible financing/EMI options. Marketing strategy should emphasize extended warranty, premium customer support, and cross-selling premium accessories.
- **Cluster B — Moderate-Value Lifestyle / Apparel Shoppers {O2, O5}:** Customers buying Clothing (₹25k–₹27k). Behavior: Frequent, trend-driven, seasonal repeat buyers. Marketing strategy should involve loyalty reward points, seasonal discount bundles, dynamic recommendation engines, and flash sales.
- **Cluster C — Mid-Tier Home & Living Segment {O4}:** Customer buying Furniture (₹30k). Behavior: Semi-frequent, home-improvement intent, heavy focus on product durability, reviews, and delivery logistics.

## Question 3: Mobile App User Behavior

### 3.1 Dataset & Feature Overview

| User ID | Daily Usage Time (min) | Login Frequency | In-App Purchase (₹) |
|---------|------------------------|-----------------|---------------------|
| U1      | 120                    | 18              | 2,000               |
| U2      | 60                     | 5               | 5,000               |

|    |     |    |       |
|----|-----|----|-------|
| U3 | 140 | 22 | 1,500 |
| U4 | 50  | 4  | 6,000 |
| U5 | 130 | 20 | 1,800 |

### 3.2 Probability-Based Membership Assignment

**Soft Membership Mechanics:** While standard Hierarchical Agglomerative Clustering (HAC) performs deterministic hard clustering, integrating probability-based soft clustering (such as Gaussian Mixture Models or Fuzzy C-Means) assigns each user a continuous membership vector  $[P(C1), P(C2), \dots, P(Ck)]$  such that  $\sum P(Ci) = 1$ .

- **Segment 1 — Highly Engaged Gamers/Browsers:** Characterized by high daily usage (120–140 min), high login frequency (18–22 times/day), but low in-app spend (₹1,500–₹2,000). Users U1, U3, and U5 exhibit high membership in this cluster (e.g., U3:  $P(\text{High-Engagement})=0.92$ ,  $P(\text{Whale})=0.08$ ).
- **Segment 2 — High-Spending 'Whales' / Monetized Users:** Characterized by lower screen time (50–60 min) and low logins (4–5 times/day), but high in-app purchases (₹5,000–₹6,000). Users U2 and U4 dominate this cluster (e.g., U4:  $P(\text{Whale})=0.95$ ,  $P(\text{High-Engagement})=0.05$ ).
- **Boundary / Hybrid Cases:** If a user displays 90 min usage and ₹3,500 spend, soft clustering captures their dual nature (e.g., 50% Engagement, 50% Monetization), avoiding artificial misclassification.

### 3.3 Advantages of Soft Clustering over Hard Clustering

- **1. Capturing Real-World Behavioral Overlap:** Consumer behavior is multi-dimensional. A gamer can be highly engaged during weekends and a high-spender during event sales. Hard clustering forces binary labels, losing nuanced behavioral signals.
- **2. Targeted Multi-Channel Monetization:** Instead of treating users rigidly, the marketing team can deliver personalized hybrid campaigns — e.g., showing rewarded video ads to users with high engagement scores while serving limited-time discount bundles to users showing growing purchase probability.
- **3. Early Detection of Behavioral Transitions (Churn/Conversion):** Tracking shifts in continuous membership probabilities over time enables early warning systems. A drop in  $P(\text{Engaged})$  from 0.85 to 0.40 flags churn risk weeks before the user becomes completely inactive.
- **4. Enhanced Granularity for Recommendation Engines:** Probabilistic weights directly feed into collaborative filtering and recommendation engines, resulting in higher conversion rates than rigid hard cluster partitions.

## Question 4: Agricultural Crop Monitoring

### 4.1 Dataset Overview

| Farm ID | Rainfall (mm) | Soil Fertility Score (1-10) | Crop Yield (tons) |
|---------|---------------|-----------------------------|-------------------|
| F1      | 300           | 8                           | 85                |
| F2      | 500           | 4                           | 70                |
| F3      | 400           | 6                           | 80                |
| F4      | 280           | 9                           | 90                |
| F5      | 520           | 3                           | 65                |

### 4.2 Impact of Feature Scale & Normalization on Clustering

**Mathematical Mechanism of Scale Bias:** In distance-based clustering algorithms (such as K-Means or Agglomerative Hierarchical Clustering using Euclidean distance  $d = \sqrt{\sum (x_i - y_i)^2}$ ), the scale of individual features drastically dictates cluster formation.

- **Raw Data Analysis (Before Normalization):** Rainfall ranges from 280 mm to 520 mm (variance ~ 11,000+), Crop Yield ranges from 65 to 90 tons (variance ~ 100), while Soil Fertility Score ranges only from 3 to 9 (variance ~ 6.5). Because Euclidean distance computes squared differences, a 200 mm difference in rainfall contributes  $(200)^2 = 40,000$  to the distance, completely dwarfing a 6-point difference in Soil Fertility ( $6^2 = 36$ ).
- **Consequence Without Normalization:** The algorithm effectively ignores Soil Fertility and Crop Yield, clustering farms solely by rainfall volume. Farms F1 and F4 group together simply because both have low rainfall (~280–300 mm), while F2 and F5 group together solely due to high rainfall (~500–520 mm).
- **Post-Normalization (Min-Max or Z-Score Scaling):** Scaling transforms all features into a uniform range [0, 1] or standard normal distribution  $N(0, 1)$ . Each feature contributes equally to the distance metric, allowing the model to uncover holistic agronomic relationships — e.g., how high soil fertility (Score 9) compensates for low rainfall (280 mm) to achieve superior crop yield (90 tons).

### 4.3 Importance for Agricultural Data Reliability & Decision-Making

- **Accurate Agronomic Profiling:** Enables researchers to identify optimal multi-variate conditions (e.g., fertile soil + moderate irrigation yield synergy) rather than one-dimensional weather splits.
- **Precision Resource Allocation:** Agricultural advisory boards can deliver customized fertilizer, seed variety, and irrigation recommendations tailored to true multi-factor land capabilities.

- **Reproducibility & Algorithm Stability:** Eliminates arbitrary measurement unit biases (e.g., changing rainfall from mm to meters or inches won't arbitrarily alter cluster boundaries).

## Question 5: Employee Performance Evaluation

### 5.1 Dataset & Silhouette Score Analysis

| Employee ID | Productivity Score | Attendance (%) | Skill Assessment Score |
|-------------|--------------------|----------------|------------------------|
| E1          | 88                 | 92             | 84                     |
| E2          | 62                 | 75             | 68                     |
| E3          | 78                 | 88             | 72                     |
| E4          | 55                 | 65             | 58                     |
| E5          | 93                 | 96             | 90                     |

| Model Configuration | Silhouette Score | Cluster Quality Interpretation                                   |
|---------------------|------------------|------------------------------------------------------------------|
| K = 2               | 0.48             | Moderate cohesion and separation; broad division.                |
| K = 3               | 0.68             | Optimal clustering; strong cohesion and clear separation (Peak). |
| K = 4               | 0.52             | Declining score; unnecessary subdivision of cohesive groups.     |

### 5.2 Justification of the Most Suitable Model (K = 3)

**Silhouette Score Evaluation:** The Silhouette Coefficient  $s(i) = [b(i) - a(i)] / \max[a(i), b(i)]$  evaluates clustering validity on a scale from -1 to +1, where  $a(i)$  is the mean intra-cluster distance and  $b(i)$  is the mean nearest-cluster distance. A higher score signifies tighter, well-isolated clusters.

- **Quantitative Comparison:** The model with K = 3 achieves the highest silhouette score of 0.68, significantly surpassing K = 2 (0.48) and K = 4 (0.52). A score of 0.68 falls into the 'strong cluster structure' bracket.
- **Data Alignment:** Examining the dataset reveals three natural employee tiers: (1) High Performers {E1, E5} with scores 84–96 across all metrics; (2) Moderate Performers {E3} with balanced metrics

around 72–88; and (3) Underperformers / Needs Support {E2, E4} with scores below 75 across the board.

### 5.3 Contribution to Effective HR Performance Evaluation & Talent Management

- **Tier 1: High Performers / Leaders {E1, E5}:** HR Action: Fast-track leadership development programs, merit-based bonuses, stock options, and mentorship roles to retain top talent.
- **Tier 2: Core Steady Contributors {E3}:** HR Action: Skill upskilling workshops, lateral growth opportunities, and performance incentives to elevate them into top performers.
- **Tier 3: Underperformers / Development Required {E2, E4}:** HR Action: Implement targeted Performance Improvement Plans (PIPs), root-cause absenteeism analysis, and tailored training bootcamps.
- **Objective & Bias-Free Appraisal:** Clustering standardizes employee evaluations using multi-dimensional metric validation, removing managerial subjectivity and appraisal recency bias.